

Triage Damaged Goods SOP (1x Episode: 4 Flagged Items)

One episode triages all **four** flagged items: each box is opened, the item is inspected on both faces, staged for a mock photo, sorted to the quarantine bin with a tag or resealed back into its box, and its condition code is logged on the code board. The episode ends after the fourth item, not after the first.

The table begins with the **flagged tote** holding four closed **item boxes** standing in a front-to-back row; where the tote stands is the episode's config (below). The **outbound tote** sits at the **back-left**, empty. The **box spot** is just left of the center of the table; every box is opened and closed there. The **clip rack** and the **used-clip cup** stand at the **front-center**, in front of the box spot, except in Config M, where they stand at the **front-right**. The **photo mat** sits on the **work spot**, just right of center, with a **corner mark** printed at its back-left corner and a **scale card** lying along its back edge. The **photo boom** stands in the **boom rest** at the **back-right**. The **quarantine bin** sits at the **mid-right**, and the **tag rack** of **quarantine tags** stands at the **back-right corner**. The **code board** with its four numbered **slots** sits at the **back-center**, with the **chip rack** of condition-code chips beside it.

The table is set up in one of three ways. Only the flagged tote moves; the outbound tote, the box spot, the photo mat, the photo boom, the quarantine bin, the tag rack, and the code board are in the same place in all three. The clip rack and the used-clip cup shift right only when the tote takes their spot.

- **Config L:** the flagged tote is at the **front-left**.
- **Config M:** the flagged tote is at the **front-center**, in front of the box spot, and the clip rack and the used-clip cup stand at the **front-right**, in front of the photo mat.
- **Config R:** the flagged tote is at the **front-right**.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions and all four items:

- **Start position:** the flagged tote starts at the front-left (**Config L**), the front-center (**Config M**), or the front-right (**Config R**). One config per episode, chosen before recording and never changed mid-episode.
- **Same-side rule:** the gripper on the flagged tote's side lifts each box out of it — the left gripper in Config L and M, the right gripper in Config R. No arm reaches across the table. In Config R the right gripper sets the box on the box spot and lets go before the left gripper presses it down; nothing is handed over in the air.

- **Fixed roles:** whichever gripper brings the box over, the left gripper holds every box on the box spot and carries every resealed box to the outbound tote, and the right gripper takes every clip, whether the clip rack is at the front-center or the front-right.
- **One box at a time:** exactly one item box is open on the table at any moment. That item is sorted and its code is logged before the next box leaves the flagged tote.
- **Front box first:** the boxes are worked in the order they stand in the flagged tote, front box first, and each item uses the code board slot matching its turn — first item to slot 1, fourth item to slot 4.
- **Look before you sort:** every item is inspected on **both faces** and staged for a **mock shot** before it is tagged, quarantined, or resealed. Nothing is sorted straight out of the box.
- **The code decides the route:** an item coded **A** is resealed into its box and goes to the outbound tote. An item coded **B** or **C** goes into the quarantine bin with a quarantine tag on it. The route always matches the code.
- **Every item gets a chip:** one condition-code chip goes into the code board slot for that item, after the mock shot and after the item has been sorted. No item is left without a chip.
- **Clips are one-use:** a seal clip that comes off a box goes straight into the used-clip cup. A box is always resealed with a **fresh** clip from the clip rack.
- **The mat is cleared:** the photo mat holds nothing but the current item and the scale card. It is empty before the next box is brought over.

The **right gripper** does the work on the right of the table and at the center: it pulls and fits the seal clips, opens and closes the lid, lifts the item to the photo mat, tips it, stages it, swings the photo boom, takes the quarantine tag, puts items into the quarantine bin, places the code chips, and in Config R brings each item box from the flagged tote to the box spot. The **left gripper** does the supporting work on the left: it brings each item box from the flagged tote to the box spot in Config L and M, presses it flat and holds it there while the right gripper works on it, and carries each resealed box to the outbound tote. The left gripper never touches the item, the photo mat, the photo boom, a quarantine tag, the quarantine bin, or a code chip.

Setup

Complete both checklists before recording starts.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.

2. The environment camera shows the whole table: the flagged tote with all four item boxes, the outbound tote, the box spot, the clip rack and the used-clip cup, the photo mat with its corner mark and scale card, the boom rest with the photo boom, the quarantine bin, the tag rack, and the code board with its four slots and the chip rack.
3. The photo mat is visible from above, so the **corner mark**, the **scale card**, and all four edges of an item lying on the mat can be seen.
4. The code board is visible clearly enough in the recording that the letter on a chip and the number on a slot can be read.
5. The item is visible clearly enough on the mat that a **scuff** the size of a fingertip can be seen on it.
6. Both arms are at home with grippers open.
7. The table is bare apart from the flagged tote with its four boxes, the outbound tote, the clip rack, the used-clip cup, the photo mat with its scale card, the boom rest with the photo boom, the quarantine bin, the tag rack, the code board with the chip rack, and robot hardware.
8. The right arm reaches the box spot, the clip rack, the used-clip cup, the whole photo mat, the boom handle, the tag rack, the quarantine bin, all four slots on the code board, and (Config R) every box in the flagged tote at the front-right without stretching. The left arm reaches the box spot, the outbound tote, and (Config L and M) every box in the flagged tote without stretching.

Materials checklist

1. The **flagged tote** sits flat and **square** to the front edge in the start zone for this episode's config. It holds exactly **four item boxes**, standing in a single front-to-back row, lids up, none stacked on another.
 - **Config L:** front-left
 - **Config M:** front-center, in front of the box spot
 - **Config R:** front-right, in front of the photo mat and clear of the quarantine bin
2. Each **item box** is closed, with its **lid** down and one **seal clip** pushed onto the middle of its front edge. No box is already open and no box has a missing clip.
3. Each item box holds exactly **one item**, lying flat, **label face up**, with nothing else in the box.
4. Of the four items, at least one has visible **damage** and at least one is **sound**, so both routes are used in the episode. The items are not sorted into any order by damage — the mixed boxes stand in the tote in whatever order the shift set them.
5. The **outbound tote** sits at the **back-left**, empty, flat, and **square** to the front edge. It is deep enough to hold four item boxes standing in a row.
6. The **box spot** is a marked rectangle just left of the center of the table, bare, and big enough for one item box with its lid folded flat open behind it.

7. The **clip rack** stands on the **supply side** — the **front-center**, in front of the box spot (Config L and R), or the **front-right**, in front of the photo mat (Config M) — holding at least **six fresh seal clips**, all the same type as the clips on the boxes, standing in a row and easy to pinch one at a time.
8. The **used-clip cup** stands beside the clip rack, empty, and wide enough to drop a clip into without aiming.
9. The **photo mat** lies on the **work spot**, just right of the center of the table, flat and **square** to the front edge. Its base does not slide when pushed by hand.
10. The photo mat has a printed **corner mark** at its **back-left corner**, big enough to see from the camera, and the **scale card** lies along the **back edge** of the mat, square to it.
11. The **photo boom** stands in the **boom rest** at the **back-right**, **mock camera head** pointing down and **boom handle** up. It swings over the photo mat and back with one steady push and holds its position.
12. The **quarantine bin** sits at the **mid-right**, empty, open-topped, low enough to set an item into without lifting the item above the bin rim by more than a hand's width, and heavy enough not to slide.
13. The **tag rack** stands at the **back-right corner**, holding at least **four quarantine tags**, standing on edge in a row, each one stiff enough to be pinched and laid flat without folding.
14. The **code board** lies at the **back-center**, flat and square to the front edge, with four **slots** printed and numbered **1, 2, 3, 4** from left to right. All four slots are empty.
15. The **chip rack** stands beside the code board, holding three labelled columns of **condition-code chips**: a column of **A** chips, a column of **B** chips, and a column of **C** chips, at least four in each.
16. Keep the table clear between the flagged tote and the box spot, and between the box spot and the outbound tote, so a box can be moved along that line without being lifted over anything.
17. Before collection, confirm by hand that:
 - a seal clip pulls off the front edge of a box in one straight pull and does not tear the box;
 - a fresh clip pushes onto a closed lid and holds it shut when the box is lifted by its sides;
 - the lid of each box folds all the way back on its hinge and lies flat behind the box;
 - an item lifted by its two side edges does not slip out of the gripper;
 - an item set on the corner mark lies fully on the mat with the scale card still showing beside it;
 - the photo boom, swung over the mat, shows the whole item in the mock camera head's view;
 - a chip dropped into a slot stays in that slot and does not slide into the next one.

Workspace layout

- **Flagged tote (start zone):** front-left (Config L), front-center in front of the box spot (Config M), or front-right (Config R) — the four closed item boxes
- **Back-left:** the outbound tote, empty, for resealed boxes
- **Box spot:** just left of the center — one item box is opened and closed here
- **Supply side:** the clip rack of fresh seal clips and the used-clip cup beside it — front-center (Config L and R) or front-right (Config M)
- **Work spot:** just right of the center — the photo mat, with the corner mark at its back-left and the scale card along its back edge
- **Back-right:** the boom rest with the photo boom, and the tag rack at the back-right corner
- **Mid-right:** the quarantine bin
- **Back-center:** the code board with slots 1 to 4, and the chip rack beside it
- **Carry line:** kept clear along the line flagged tote → box spot → outbound tote

Arm assignments

- **Left gripper:** lifts the next item box out of the flagged tote and sets it on the box spot (Config L and M), presses it flat and holds it there while the right gripper works on it, and carries each resealed box to the outbound tote.
- **Right gripper:** in Config R, lifts the next item box out of the flagged tote at the front-right and sets it on the box spot; pulls the seal clip and drops it in the used-clip cup, opens and closes the lid, lifts the item to the photo mat, tips it to show its other face, stages it on the corner mark, swings the photo boom over and back, takes a quarantine tag, sets quarantined items and their tags into the quarantine bin, puts sound items back in their box, fits a fresh clip, and drops the condition-code chip into its slot.

Vocabulary

- **Square:** the front edge of a part lines up with the front edge of the table, so neither end of the part sits nearer the front.
- **Flagged tote:** the tote holding the four item boxes to be triaged. It is the **start zone** — front-left (**Config L**), front-center (**Config M**), or front-right (**Config R**). One per episode, chosen before recording and never changed mid-episode. It is never lifted.
- **Outbound tote:** the empty tote at the back-left that resealed boxes are set into. It is never lifted.
- **Item box:** the shallow box one item lies in. It has a **lid** on a hinge at its back edge and a **seal clip** on its front edge.
- **Lid:** the hinged top of an item box. Open means folded all the way back so it lies flat on the table behind the box.

- **Seal clip:** the stiff clip that holds a box lid shut, pushed onto the middle of the box's front edge.
- **Clip rack:** the holder on the supply side with fresh seal clips standing in a row.
- **Used-clip cup:** the cup beside the clip rack that every clip pulled off a box is dropped into. No clip in this cup is used again in the episode.
- **Supply side:** where the clip rack and the used-clip cup stand — the front-center (**Config L** and **R**) or the front-right (**Config M**). The right gripper takes clips there in all three.
- **Fresh clip:** a clip taken from the clip rack. A clip taken from the used-clip cup or picked off the table is not a fresh clip.
- **Box spot:** the marked rectangle just left of center where one item box sits while it is open. Only one box is on it at a time.
- **Item:** the single object inside an item box. It is what is inspected, photographed, and sorted.
- **Label face:** the face of the item that carries its printed label. It is the face that lies up in the box.
- **Side edges:** the two short edges of the item. They are the only part of the item a gripper pinches.
- **Damage:** anything on the item that was not there when it was packed — a scuff, a dent, a crack, a split seam, a tear, a loose or rattling part, or any wet or sticky patch.
- **Sound:** no damage is visible on either face of the item.
- **Light damage:** the item is marked but whole — a scuff, a dent, a crease, or a torn outer wrap, with no crack, no split, and nothing loose inside.
- **Broken:** the item is not whole — a crack, a split seam, a piece off or loose inside, or anything wet or leaking.
- **Photo mat:** the flat mat on the work spot the item is laid on for inspection and for the mock shot. It is never lifted.
- **Corner mark:** the printed mark at the back-left corner of the photo mat. The back-left corner of the item is set on it.
- **Scale card:** the marked card lying along the back edge of the photo mat, so the mock shot shows the item next to something of known size. It stays on the mat all episode.
- **Tipped:** the item has been turned over on the mat by pushing its far edge down toward the front edge, so it lands on its other face without being lifted or twisted in the air.
- **Staged:** the item lies flat on the photo mat with its back-left corner on the corner mark, its edges square to the mat, the scale card still showing along the back edge, and the face named in Step 5 facing up.
- **Photo boom:** the arm at the back-right with the **mock camera head** on its end. It is picked up only by its **boom handle**.
- **Boom handle:** the grip on the photo boom arm. It is the only part of the boom a gripper touches.

- **Boom rest:** the cradle at the back-right the photo boom sits in when it is not over the photo mat.
- **Mock shot:** the photo boom is held still over the staged item for 2 seconds with nothing shading the item and nothing but the item and the scale card on the mat. No picture is actually taken.
- **Quarantine bin:** the open bin at the mid-right that items with light damage or broken items are set into.
- **Quarantine tag:** the stiff card laid on top of a quarantined item in the bin, so the item in the bin can be told apart from anything else in it.
- **Tag rack:** the holder at the back-right corner the quarantine tags stand in.
- **Code board:** the board at the back-center with four numbered slots, one per item, in the order the items are worked.
- **Slot:** one numbered place on the code board. Slot 1 is for the first item worked, slot 4 for the fourth.
- **Chip rack:** the holder beside the code board with three labelled columns of condition-code chips.
- **Condition code:** the letter that says what was found on the item. **A** means sound. **B** means light damage. **C** means broken.
- **Logged:** one chip lies in the slot for that item, letter side up, and the letter can be read from above.
- **Cleared mat:** nothing is on the photo mat but the scale card.

Steps

Work the four item boxes in the order they stand in the flagged tote, **front box first**. Run Steps 1–7 in order on one box, then go back to Step 1 for the next box. After the fourth box has been logged in Step 7, end the episode with Step 8.

Only the first pick in Step 1 and the clip work in Steps 2.2 and 6 depend on the config. In Config L and M the **left gripper** lifts the box out of the flagged tote; in Config R the **right gripper** lifts it out of the tote at the front-right, sets it on the box spot, and lets go before the left gripper presses it down. In Config M the clip rack and the used-clip cup stand at the front-right instead of the front-center. Every other line is the same in all three configs.

Step 1: Bring the next item box to the box spot

Goal: one closed item box sits square on the box spot, held flat by the left gripper.

Look where the flagged tote is before reaching for the box.

- **IF the flagged tote is at the front-left (Config L):** with the **left gripper**, pinch the **front-most item box** in the tote across its two sides, lift it straight up out of the tote, carry it low

over the table straight back and right to the **box spot**, and set it down flat, **square** to the front edge, with its lid up and its **seal clip** facing the front edge.

- **IF the flagged tote is at the front-center (Config M):** with the **left gripper**, pinch the **front-most item box** across its two sides, lift it straight up, carry it low over the table straight back to the **box spot**, and set it down flat, **square** to the front edge, lid up, seal clip facing the front edge.
- **IF the flagged tote is at the front-right (Config R):** with the **right gripper**, pinch the **front-most item box** across its two sides, lift it straight up, carry it low over the table back and left to the **box spot**, set it down flat, **square** to the front edge, lid up, seal clip facing the front edge, and let go. The right gripper is clear of the box before the left gripper touches it.

Then, in all three:

- With the **left gripper**, press down on the **left side** of the box and hold it against the table.
- Keep the **left gripper** there through Steps 2 and 6. It comes off only when the box is finished.

Check: exactly one box is on the box spot, it sits flat and square with its seal clip toward the front edge, and it does not slide when the left gripper presses. The photo mat is a **cleared mat**. If the box is crooked, straighten it with the **left gripper** before pressing it down. If a second box is on the box spot, put the extra one back in the flagged tote with the gripper that brought it before going on.

Expected state: one closed box is held on the box spot, the flagged tote holds the boxes not yet worked, the photo mat holds only the scale card, and the right gripper is empty.

Step 2: Unseal and open the box

Goal: the box lid lies flat open behind the box, the used clip is in the used-clip cup, and the item is showing.

- With the **left gripper**, keep pressing the box down.
- The **right gripper** does all the clip and lid work. The lid is never forced and the item is never shaken or tipped out.

2.1 Pull the seal clip off

- With the **right gripper**, pinch the **seal clip** on the front edge of the box.
- With the **right gripper**, pull the clip straight **toward the front edge** in one steady pull until it comes off the box. Do not twist it and do not lift it upward off the edge.
- If the clip does not come off in one pull, pinch it again with the **right gripper** and pull the same way. Do not lever the lid open with the clip still on.

2.2 Drop the used clip in the cup

- With the **right gripper**, carry the clip to the **used-clip cup** and let go over the cup. **IF Config L or R:** the cup is at the front-center. **IF Config M:** it is at the front-right, in front of the photo mat.
- The **right gripper** is empty before it touches the lid.

2.3 Open the lid

- With the **right gripper**, pinch the **front edge of the lid** and lift it up and back over its hinge.
- With the **right gripper**, lay the lid flat on the table **behind the box** and let go.
- The lid stays flat behind the box until Step 6. Do not stand it up and do not rest it on the box.

Check: the lid lies flat behind the box, the used clip is in the used-clip cup, and the item lies flat in the box **label face up**, undisturbed. If the item has shifted or turned over while the lid was opened, leave it as it is — Step 3 lifts it out either way. If the used clip landed on the table instead of in the cup, pick it up with the **right gripper** and drop it in the cup before going on.

Expected state: one open box is held on the box spot with its lid flat behind it, the item is showing, the used-clip cup holds the clip, and the right gripper is empty.

Step 3: Lift the item onto the photo mat

Goal: the item lies flat on the photo mat, label face up, ready to be inspected.

- With the **left gripper**, keep pressing the box down.
- The **right gripper** pinches the item by its **side edges** only. It never pinches the item across a face, and it never squeezes a **crack**, a **split seam**, or a soft or wet patch.
- With the **right gripper**, pinch the item across its two **side edges** and lift it straight up out of the box.
- Keep the item level in the **right gripper**. Do not tilt it and do not turn it over in the air.
- With the **right gripper**, carry the item right to the **photo mat**, keeping it low over the table.
- With the **right gripper**, set the item down flat on the mat, **label face up**, clear of the **scale card**, and let go.

Check: the item lies flat on the mat, label face up, with the scale card still lying along the back edge and not under the item. The item box on the box spot is empty. If the item landed on the scale card or pushed it out of place, lift the item with the **right gripper**, put the scale card back along the back edge, and set the item down clear of it.

Expected state: the item lies flat on the photo mat label face up, the scale card is along the back edge, the box is open and empty on the box spot, and the right gripper is empty.

Step 4: Inspect both faces

Goal: both faces of the item have been looked at on the mat, and what is on them is known.

- With the **left gripper**, keep pressing the box down.
- The item is turned by **tipping it on the mat**, never by lifting and twisting it in the air.

4.1 Look at the label face

- Look at the whole of the item's **label face** and at all four of its edges for **damage**.
- Keep the **right gripper** and the arm to the **side** of the mat, not over the item, so nothing shades the item or hides it from the camera.

4.2 Tip the item and look at the other face

- With the **right gripper**, press down on the item's **back edge** and push it toward the **front edge** until the item rolls over onto its other face and lies flat on the mat.
- Let go with the **right gripper** and move the arm back to the side of the mat.
- Look at the whole of that face and at all four edges for **damage**.
- Do not lift the item off the mat and do not twist the gripper to turn it.

4.3 Decide the condition code

- If no damage shows on either face, the code is **A**.
- If the item is marked but whole — **light damage** — the code is **B**.
- If the item is not whole — **broken** — the code is **C**.
- Hold that code for the rest of this item. It decides both the mock shot in Step 5 and the route in Step 6.

Check: both faces of the item have been on top and have been looked at with the arm clear of the item. The item is still on the mat and the scale card is still along the back edge. If the item slid off the mat while it was tipped, put it back on the mat with the **right gripper** and tip it again. If the item is now on its label face, that is fine — Step 5 sets the face that goes up.

Expected state: the item lies flat on the mat, both faces have been looked at, one condition code is decided, and the right gripper is empty.

Step 5: Stage the item and take the mock shot

Goal: the item is **staged** on the corner mark and one **mock shot** is taken over it.

- With the **left gripper**, keep pressing the box down.
- The **right gripper** must be empty of everything before it takes the **boom handle**.

5.1 Set the face that goes up

- If damage shows on **one face only**, turn that face up: with the **right gripper**, tip the item on the mat as in 4.2 until the damaged face is up.

- If damage shows on **both faces**, or on **neither face**, turn the **label face** up the same way.
- Do not lift the item to turn it. Tip it on the mat.

5.2 Stage the item on the corner mark

- With the **right gripper**, slide the item across the mat until its **back-left corner** sits on the **corner mark**.
- With the **right gripper**, square the item to the mat, so its back edge runs parallel to the mat's back edge.
- Leave the **scale card** where it is, along the back edge, showing beside the item and not under it.
- Let go with the **right gripper** and move the arm to the **side** of the mat.

5.3 Swing the boom over and hold

- With the **right gripper**, pinch the **boom handle** and swing the **photo boom** out of the **boom rest** and over the photo mat, so the **mock camera head** sits above the item.
- Hold the boom still there for **2 seconds**, with the arm to the side of the mat so nothing shades the item or hides it from the camera.
- With the **right gripper**, swing the boom back and set it in the **boom rest**.

Check: the item was **staged** — back-left corner on the corner mark, square to the mat, scale card showing along the back edge, and the face named in 5.1 up — before the boom came over, and the boom was held still for 2 seconds. The boom is back in its rest and the right gripper is empty. If the item moved while the boom was over it, stage it again with the **right gripper** and repeat 5.3 once. Do this repeat **at most once**, then go on to Step 6 whatever the second shot looks like.

Expected state: one mock shot has been taken over the staged item, the photo boom sits in its rest, the right gripper is empty, and the item is still on the corner mark.

Step 6: Sort the item

Goal: the item has gone to the route its code calls for — quarantined with a tag, or resealed into its box and put in the outbound tote.

- With the **left gripper**, keep pressing the box down until this step says to let go.
- The route always matches the code decided in 4.3. **A** goes to 6.1. **B** or **C** goes to 6.2.
- **IF Config L or R:** the fresh clip comes from the clip rack at the front-center. **IF Config M:** it comes from the clip rack at the front-right, in front of the photo mat.

6.1 Code A — reseal the item into its box

- With the **right gripper**, pinch the item by its **side edges**, lift it off the mat, carry it low over the table, and set it flat in the open **item box**, **label face up**.

- With the **right gripper**, pinch the **far edge of the lid** where it lies behind the box, lift it up and over the hinge, and lower it flat onto the box.
- With the **right gripper**, take one **fresh clip** from the **clip rack**.
- With the **right gripper**, push the fresh clip straight onto the middle of the box's **front edge** until it holds the lid shut, and let go.
- Open the **left gripper** and lift it off the box.
- With the **left gripper**, pinch the closed box across its two sides, lift it off the box spot, carry it low over the table to the **outbound tote** at the back-left, and stand it in the tote behind the boxes already there.
- Go to Step 7.

6.2 Codes B and C — quarantine the item with a tag

- With the **right gripper**, pinch the item by its **side edges**, lift it off the mat, and carry it low over the table to the **quarantine bin** at the mid-right.
- With the **right gripper**, lower the item into the bin until it is just above the bottom or just above what is already in the bin, then let go. Do not drop it in from above the bin rim and do not throw it.
- With the **right gripper**, pinch one **quarantine tag** from the **tag rack** at the back-right corner.
- With the **right gripper**, lower the tag into the bin and lay it flat **on top of that item**, then let go. One tag per quarantined item.
- Open the **left gripper** and lift it off the box.
- With the **right gripper**, pinch the **far edge of the lid**, lift it over the hinge, and lower it flat onto the **empty** box.
- With the **right gripper**, take one **fresh clip** from the **clip rack** and push it onto the middle of the box's front edge until it holds the lid shut, and let go.
- With the **left gripper**, pinch the closed empty box across its two sides, lift it off the box spot, and stand it in the **outbound tote** behind the boxes already there.
- Go to Step 7.

Check: the route matches the code. For **A**, the item is back in its box, the lid is shut, a **fresh** clip holds it, and the box stands in the outbound tote. For **B** or **C**, the item lies in the quarantine bin with one tag flat on top of it, and its empty box is shut with a fresh clip and standing in the outbound tote. The box spot is bare and the photo mat is a **cleared mat** apart from the scale card. If the lid will not shut because the item sits proud, lift the item with the **right gripper**, lay it flat again, and shut the lid. If a clip was taken from the used-clip cup instead of the clip rack, pull it off with the **right gripper**, drop it back in the cup, and fit one from the rack.

Expected state: the item is on its route, the box spot is bare, the photo mat holds only the scale card, both grippers are empty, and the code for this item is still to be logged.

Step 7: Log the condition code

Goal: one chip showing this item's code lies in the code board slot for this item.

- With the **right gripper**, pinch one chip from the **chip rack** column matching the code decided in 4.3 — **A**, **B**, or **C**.
- With the **right gripper**, carry the chip to the **code board** at the back-center and lower it into the **slot** for this item: slot **1** for the first item worked, **2** for the second, **3** for the third, **4** for the fourth.
- With the **right gripper**, let the chip go flat in the slot, **letter side up**, and lift the arm clear.

Check: the chip is **logged** — it lies in the right slot, letter side up, and the letter reads from above. The letter matches the route just taken: an **A** chip only where the item went to the outbound tote in its box, a **B** or **C** chip only where the item went into the quarantine bin. Slots fill left to right with no slot skipped. If the chip lands in the wrong slot or on its face, pick it up with the **right gripper** and set it down again in the right slot letter side up. If a chip lands on the table, pick it up with the **right gripper** and put it in its slot.

Expected state: the code board shows one chip for every item worked so far, the box spot is bare, the photo mat holds only the scale card, and both grippers are empty.

Next: if item boxes are still in the flagged tote, go back to **Step 1** for the next box. If the flagged tote is empty and four chips are on the code board, go to **Step 8**.

Step 8: End the episode

Goal: recording ends with all four items triaged, sorted, and logged.

1. Confirm the end state:

- the flagged tote is empty;
- four boxes stand in the outbound tote, each with its lid shut and a fresh clip on its front edge;
- every item coded **B** or **C** lies in the quarantine bin with one tag flat on top of it;
- every item coded **A** is inside its own box in the outbound tote;
- four chips lie on the code board, one in each slot from 1 to 4, letter side up;
- the used-clip cup holds four used clips and none are on the table;
- the photo mat is a **cleared mat** with the scale card along its back edge;
- the photo boom sits in the boom rest and the box spot is bare.

2. Return both arms home with grippers open. Homing is the last thing the arms do.

3. Stop recording.

After the episode: reset the workspace

All reset work happens with recording off.

Between episodes

1. Lift the four boxes out of the outbound tote. Pull the clip off each one and drop it in the used-clip cup.
2. Take the quarantined items out of the quarantine bin and take the tags off them. Set the tags aside to be restocked.
3. Put one item back in each box, lying flat and **label face up**, keeping the same mix of damaged and sound items the set was built with. Replace any item that has picked up new damage during the episode with a spare in the same condition, so the mix stays the same.
4. Shut each lid and push one clip from the clip rack onto its front edge. Stand the four boxes in the **flagged tote** in a front-to-back row, lids up, clips toward the front edge, and set the tote in the start zone for the next episode's config — front-left (Config L), front-center in front of the box spot (Config M), or front-right (Config R).
5. Empty the used-clip cup and restock the **clip rack** to at least six fresh clips. Stand both at the front-center, or at the front-right if the next episode is Config M.
6. Take the four chips off the **code board** and put them back in their columns on the chip rack. Leave all four slots empty.
7. Stand the quarantine tags back in the **tag rack**, at least four of them, and leave the **quarantine bin** empty.
8. Wipe the **photo mat**, lay the **scale card** back along its back edge, and set the mat **square** on the work spot. Set the photo boom in the **boom rest**.
9. Run both Setup checklists before the next episode.

At the end of the set

1. Inspect the parts. Replace any **seal clip** that no longer holds a lid shut, and any **item box** with a torn hinge, a crushed corner, or a front edge a clip will not grip.
2. Replace any **quarantine tag** that is bent, soft, or no longer stands up in the tag rack, and any **condition-code chip** whose letter has worn off.
3. Check that the **photo boom** still swings over the mat and holds its position without drooping. If it droops, flag the boom and stop collection on this station.
4. Check that the **code board** slots still hold a chip without letting it slide, and that the **quarantine bin** still sits without sliding on the table.
5. Leave four closed boxes standing in the flagged tote, the outbound tote empty, the quarantine bin empty, the box spot bare, all four code board slots empty, the chip rack and tag rack stocked, the clip rack stocked, the used-clip cup empty, the scale card on the mat, and the photo boom in its rest.

SOP violations

Things that break this SOP and that reviewers look for in the side-by-side review tool.

How to record a violation in review

For every violation seen in a recorded episode, record:

- the **start timestamp** in the video;
- the **violation name** from the list below; and
- the **SOP rule broken**, including the step number.

The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. An episode may contain multiple violations, and the same violation may be tagged more than once if it happens on more than one item; tag each separately. Retain the episode in training data with its violation tags. Do not delete a recorded episode solely because it contains a violation.

Violations

Note on the start position: the violations below were written for Config L (flagged tote at the front-left, clip rack and used-clip cup at the front-center). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the flagged tote is not in the start zone for the config; a gripper reaches across the table for an item box; in Config R the right gripper holds the box on the box spot or carries it to the outbound tote, or the left gripper reaches to the front-right for a box; the clip rack and used-clip cup are not on the supply side for the config; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the same-side rule (the left gripper lifts each box out of the flagged tote in Config L and M, the right gripper in Config R and sets it on the box spot for the left gripper to hold; no arm reaches across the table; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the flagged tote is before the first reach, then follow that config's IF lines through Steps 1, 2 and 6.

Violation: Item box not held on the box spot

- **Visible cue:** the left gripper is off the box while the right gripper pulls the clip, opens the lid, lifts the item out, puts it back, shuts the lid, or fits a fresh clip, and the box slides, lifts, tips, or is chased across the table.
- **SOP rule broken:** Steps 1, 2, 3, 4, 5 and 6 (the left gripper presses the box down by its left side on the box spot and holds it there until Step 6 says to let go).
- **Coaching note:** left gripper on the box first, and it stays there until the box is finished.

Violation: More than one box open, or boxes worked out of order

- **Visible cue:** two item boxes are on the box spot or open at once; or the left gripper takes a box that is not the front-most one in the flagged tote; or a second box is opened before the first item has been sorted and its chip is on the code board.
- **SOP rule broken:** Steps 1 and 7 (one box at a time on the box spot, front box first, and the next box is not brought over until the current item is sorted and logged).
- **Coaching note:** front box, all the way through to its chip, then the next box.

Violation: Box opened the wrong way

- **Visible cue:** the right gripper twists the seal clip, lifts it upward off the edge, levers the lid open with the clip still on, tears the box, or opens the lid before the clip is off; or the lid ends up standing up, resting on the box, folded under it, hanging over the front edge, torn off its hinge, or laid on the photo mat or in a tote instead of flat on the table behind the box.
- **SOP rule broken:** Steps 2.1 and 2.3 (pull the clip straight toward the front edge in one steady pull, never lever the lid with the clip on, then lift the lid over its hinge and lay it flat behind the box, where it stays until Step 6).
- **Coaching note:** clip off first, straight toward the front, then the lid all the way back and flat.

Violation: Used clip not put in the used-clip cup

- **Visible cue:** the right gripper drops or lays the pulled clip on the table, in the item box, in a tote, on the photo mat, or on the floor; or is still holding it when it takes the lid; or a clip left over from an earlier item is still lying on the table at the end of the episode.
- **SOP rule broken:** Step 2.2 (carry the clip to the used-clip cup and let go over the cup, with an empty gripper before the lid is touched).
- **Coaching note:** clip goes in the cup before the hand goes anywhere else.

Violation: Item shaken, tipped, or poured out of the box

- **Visible cue:** the box is tilted, shaken, or turned over to get the item out; or the item slides out of the box onto the table or the mat instead of being lifted by the right gripper.
- **SOP rule broken:** Step 3 (the right gripper pinches the item by its side edges and lifts it straight up out of the box).
- **Coaching note:** lift the item out. The box never tips.

Violation: Item gripped on a face or on the damage

- **Visible cue:** a gripper pinches the item across a face instead of its two side edges, squeezes a crack, a split seam, or a wet patch, or presses on the damaged area while sliding or tipping the item.
- **SOP rule broken:** Steps 3, 4, 5 and 6 (the right gripper pinches the item by its side edges only and never squeezes a crack, a split seam, or a soft or wet patch).
- **Coaching note:** side edges only. Never take hold of the broken part.

Violation: Only one face inspected

- **Visible cue:** the item goes from the box to the mock shot, the bin, or its box without ever being tipped over, so one face is never on top; or the arm stands over the item through the look, shading it or hiding it from the camera.
- **SOP rule broken:** Step 4 (look at the label face, tip the item, and look at the other face, with the arm kept to the side of the mat both times).
- **Coaching note:** both faces get a look, arm to the side each time. Nothing is sorted off one face.

Violation: Item lifted and twisted instead of tipped

- **Visible cue:** the right gripper lifts the item off the mat and rolls the wrist to turn it over in the air, instead of pressing its back edge down toward the front edge to roll it on the mat.
- **SOP rule broken:** Steps 4.2 and 5.1 (turn the item by tipping it on the mat; do not lift it and twist it in the air).
- **Coaching note:** push the far edge down and let it roll on the mat. The wrist stays out of it.

Violation: Wrong face up for the mock shot

- **Visible cue:** damage shows on one face only and the item is staged with the clean face up; or the item is clean on both faces, or damaged on both, and it is staged with the label face down.
- **SOP rule broken:** Step 5.1 (damage on one face only goes up; otherwise the label face goes up).
- **Coaching note:** the shot has to show what was found. Damaged face up, or label face if there is nothing to show.

Violation: Item not staged on the corner mark

- **Visible cue:** the item sits away from the corner mark, hangs off an edge of the mat, or lies skewed to the mat when the boom comes over.
- **SOP rule broken:** Step 5.2 (slide the item until its back-left corner is on the corner mark and square it to the mat before the boom comes over).
- **Coaching note:** corner on the mark, edges square, then the boom.

Violation: Scale card covered, moved, or off the mat

- **Visible cue:** the item is set down on top of the scale card, the card is pushed off the back edge, knocked off the mat, moved to another part of the mat, or picked up by a gripper.
- **SOP rule broken:** Steps 3 and 5.2 (the scale card stays along the back edge of the mat and shows beside the item, never under it).
- **Coaching note:** the card is part of the shot. Put the item beside it, never on it.

Violation: Mock shot skipped or too short

- **Visible cue:** the item is sorted without the right gripper ever taking the boom handle; or the boom swings over and back in less than 2 seconds; or the arm stands over the item through the whole hold, shading it or hiding it from the camera.
- **SOP rule broken:** Step 5.3 (swing the boom over the staged item, hold it still for 2 seconds, and keep the arm to the side of the mat).
- **Coaching note:** every item gets one shot. Arm to the side, hold for two.

Violation: Photo boom not put back in its rest

- **Visible cue:** the boom is left hanging over the mat, resting on the table or the mat, or still in the right gripper when the item is lifted; or the right gripper grabs the mock camera head instead of the boom handle.
- **SOP rule broken:** Step 5.3 (pinch the boom handle only, and swing the boom back into the boom rest after the hold).
- **Coaching note:** handle only, and the boom goes home before the item moves.

Violation: Route does not match the code

- **Visible cue:** an item with visible damage is put back in its box and sent to the outbound tote; or an item with no damage on either face is put into the quarantine bin; or the chip letter logged in Step 7 does not match the route the item actually took.
- **SOP rule broken:** Steps 6.1, 6.2 and 7 (code **A** is resealed into its box and goes to the outbound tote; codes **B** and **C** go into the quarantine bin with a tag; the chip matches the route).
- **Coaching note:** say the code out loud before you move the item, then move it where that code goes.

Violation: Quarantine tag skipped or misplaced

- **Visible cue:** an item is put in the quarantine bin and no tag follows it; or the tag is laid on the table, on the mat, on the bin rim, in a tote, or in the bin but not on that item; or two items in the bin share one tag; or two tags go in for one item.
- **SOP rule broken:** Step 6.2 (take one quarantine tag from the tag rack and lay it flat on top of that item in the bin — one tag per quarantined item).

- **Coaching note:** the item goes in, then its own tag goes flat on top of it. One for one.

Violation: Item dropped into the quarantine bin

- **Visible cue:** the right gripper opens above the bin rim and lets the item fall, throws or tosses it in, or lets go of it while it is still tipped or held by a face.
- **SOP rule broken:** Step 6.2 (lower the item into the bin until it is just above the bottom or just above what is already there, then let go).
- **Coaching note:** take it all the way down before you open the gripper. Damaged goods do not get dropped.

Violation: Box not resealed, or resealed with a used clip

- **Visible cue:** a box goes to the outbound tote with its lid open, with no clip on its front edge, with the lid only laid on, or with a clip taken from the used-clip cup or picked up off the table instead of from the clip rack.
- **SOP rule broken:** Steps 6.1 and 6.2 (shut the lid and push a **fresh** clip from the clip rack onto the middle of the front edge until it holds the lid shut — every box, whether the item went back in or not).
- **Coaching note:** every box gets its lid shut and a fresh clip, even the empty ones.

Violation: Box put back in the flagged tote or left out

- **Visible cue:** a finished box is stood back in the flagged tote, left on the box spot, set on the table, put in the quarantine bin, or laid on the photo mat instead of standing in the outbound tote.
- **SOP rule broken:** Steps 6.1 and 6.2 (the left gripper carries the closed box off the box spot and stands it in the outbound tote behind the boxes already there).
- **Coaching note:** finished boxes only ever go to the outbound tote. The flagged tote only empties.

Violation: Condition code not logged

- **Visible cue:** the next box is brought to the box spot, or the episode ends, with no chip in the slot for the item just sorted; or the code board shows fewer than four chips at the end of the episode.
- **SOP rule broken:** Step 7 (drop one chip into the code board slot for that item before going back to Step 1 or on to Step 8).
- **Coaching note:** the item is not finished until its chip is in its slot. Chip, then the next box.

Violation: Chip in the wrong slot or unreadable

- **Visible cue:** a chip goes into a slot out of turn, a slot is skipped, two chips go into one slot, or a chip lies letter side down, on its edge, half out of the slot, or on the board between slots.

- **SOP rule broken:** Step 7 (slot 1 for the first item worked through slot 4 for the fourth, one chip per slot, lying flat letter side up so the letter reads from above).
- **Coaching note:** slots fill left to right, one chip each, letter facing the camera.

Violation: Chip taken before the mock shot

- **Visible cue:** the right gripper takes a chip from the chip rack, or puts one in a slot, before the photo boom has been swung over that item and held for 2 seconds.
- **SOP rule broken:** Steps 5.3, 6 and 7 (the mock shot is taken and the item is sorted before the chip is picked up).
- **Coaching note:** look, shoot, sort, then log. The chip is always last.

Violation: Mat or box spot not cleared before the next item

- **Visible cue:** the left gripper brings the next box to the box spot while the previous item, its empty box, a used clip, a spare tag, or a chip is still on the photo mat or on the box spot.
- **SOP rule broken:** Steps 1, 6 and 7 (the box spot is bare and the photo mat holds only the scale card before the next box is set down).
- **Coaching note:** clear the mat and the spot, then bring the next box over.

Violation: Wrong arm used

- **Visible cue:** an action assigned to one gripper is done by the other, including the left gripper pulling a clip, opening or shutting a lid, touching the item, staging it, swinging the photo boom, taking a tag, reaching into the quarantine bin, or placing a chip; or the right gripper lifting a box out of the flagged tote, holding the box down on the box spot, or carrying a box to the outbound tote.
- **SOP rule broken:** Steps 1–7 (the right gripper does the clip, lid, item, boom, tag, bin, and chip work; the left gripper fetches, holds, and puts away the boxes).
- **Coaching note:** right gripper for everything from the clip to the chip, left gripper for the boxes. It does not swap.

Violation: Part dropped or knocked off its spot

- **Visible cue:** either gripper drops an item, a box, a clip, a tag, or a chip on the table or the floor; or tips, drags, or pushes the flagged tote, the outbound tote, the clip rack, the used-clip cup, the photo mat, the quarantine bin, the tag rack, the code board, or the chip rack off its spot; or knocks the chips already on the code board out of their slots.
- **SOP rule broken:** Steps 1–7 (every tote, rack, cup, mat, bin, and board stays on its spot through the whole episode, and nothing leaves a gripper except where the step says).
- **Coaching note:** work slower and lower over the table, and keep each carry short.

Violation: Wrong episode ending

- **Visible cue:** recording stops before the empty flagged tote, the four resealed boxes, the tagged items in the bin, and the four chips are confirmed; or before all four items have been worked; or an arm is not home, a gripper is closed, or an arm does something else after homing.
- **SOP rule broken:** Step 8 (confirm the end state, return both arms home with grippers open as their final action, then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Non-violation failures

Failures not caused by how the task was run are system issues. Log and discard the episode rather than tagging them as SOP violations.

- Recording stops or pauses during the episode.
- A camera drops frames or loses its feed.
- An arm or gripper fails, drifts, or reports a motor error.
- An item box arrives with a torn hinge or a crushed front edge, so no clip will hold its lid shut.
- A seal clip arrives split or sprung, so it falls off a closed lid on its own.
- An item box arrives empty, or holds more than one item.
- The photo boom will not hold its position over the mat and droops onto the item.
- The scale card arrives creased or curled so it will not lie flat along the back edge of the mat.
- A quarantine tag arrives soft or folded and cannot be laid flat on an item.
- A condition-code chip arrives with no letter on it, or the code board slots are too shallow to hold a chip.
- The item is damaged so badly that it cannot be lifted by its side edges in one piece.

Annotation subtasks (from SOP)

1. Lift the front item box out of the flagged tote and set it on the box spot with the left gripper
2. Press the box flat and hold it with the left gripper
3. Pull the seal clip off the front edge with the right gripper
4. Drop the used clip in the used-clip cup
5. Open the lid and lay it flat behind the box
6. Lift the item out by its side edges and set it on the photo mat
7. Look at the label face with the arm clear of the item
8. Tip the item over on the mat and look at the other face
9. Turn the face that goes up and slide the item onto the corner mark
10. Swing the photo boom over the item, hold it for two seconds, and swing it back
11. Set a code A item back in its box, label face up

12. Set a code B or C item down into the quarantine bin
13. Lay one quarantine tag flat on the item in the bin
14. Shut the lid and push a fresh clip onto the front edge
15. Stand the closed box in the outbound tote with the left gripper
16. Take the matching chip from the chip rack and drop it in this item's slot
17. Go back to the flagged tote for the next box
18. Confirm the end state, return both arms home, and end the episode